

ABDULLAH KHALIL

Full-Stack Developer | +92 317 767 8576 | abdullahkhalil8198@gmail.com
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SUMMARY

BS Artificial Intelligence student at NUST (2024–2028) with professional experience as a Full-Stack Developer at Rewild Mobile (UK) and an active freelance track record since 2023. Builds production-grade web applications using the MERN stack, REST APIs, TypeScript, and cloud deployment pipelines. Integrates AI/ML features including RAG pipelines, semantic search, and NLP models into full-stack products. Delivered 3+ major projects on schedule. Harvard certified: CS50AI, CS50P.

EDUCATION

BS Artificial Intelligence

Sep 2024 – Jun 2028

NUST — National University of Sciences and Technology, Islamabad

- Relevant coursework: Machine Learning, AI Programming, Web Technologies, Data Structures and Algorithms, Software Engineering

CS50AI — Artificial Intelligence | Harvard University (edX)

Jan 2024

CS50P — Python Programming | Harvard University (edX)

Jul 2024

SKILLS

Machine Learning / AI: supervised learning, classification, regression, scikit-learn, TensorFlow, PyTorch, feature engineering, hyperparameter tuning, cross-validation, stacking ensembles, SHAP, joblib, Pandas, NumPy

Deep Learning / NLP: XLM-RoBERTa, HuggingFace Transformers, fine-tuning, fp16, gradient checkpointing, TF-IDF, sentiment analysis, multilingual NLP

LLMs / Agents / RAG: LangChain, LangGraph, OpenAI API, Google Gemini, Pinecone, FAISS, RAG pipelines, multi-agent systems, prompt engineering, vector search, Tavily MCP, Chainlit

MLOps / DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, drift detection, model versioning, model deployment, microservices

Backend / APIs: FastAPI, REST API, Node.js, Express.js, Django, Pydantic, JWT, Spring Boot

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, WebSocket

Databases: MongoDB, PostgreSQL, MySQL, SQLAlchemy, Mongoose, SQLite

Languages: Python, JavaScript, TypeScript, Java, C/C++

AI Algorithms: CSP, backtracking search, MRV heuristic, Arc Consistency (AC-3), forward checking, A*, BFS, DFS

EXPERIENCE

Full-Stack Developer

Remote, Pakistan

Rewild Mobile Ltd (UK)

Jan 2025 – Present

- Built and deployed responsive web interfaces for institutional platforms using React.js and Tailwind CSS, reducing UI bug reports by 20%
- Designed and consumed RESTful APIs integrating dynamic frontend components with backend microservices
- Improved database query speed by 30% through optimized schema design and data-handling strategies in PostgreSQL and MongoDB
- Collaborated in Agile sprints with cross-functional teams, delivering all assigned features on schedule

Freelance Full-Stack Developer

Remote

Self-Employed

2023 – Present

- Designed and developed 5+ client websites with mobile-first, fully responsive layouts using React.js and Tailwind CSS
- Delivered pixel-perfect UIs with REST API integrations, managing multiple projects simultaneously and meeting all deadlines

PROJECTS

SmartPulse — AI Business Intelligence SaaS Platform | Python, scikit-learn, FastAPI, React, Docker, Kubernetes, GitHub Actions

2026 – Present

- Built full ML pipeline: Linear Regression and Random Forest for sales forecasting, Logistic Regression, SVM, and MLP for churn classification, K-Means for customer segmentation, and GDA for anomaly detection
- Served all models via FastAPI microservices with Pydantic validation, Swagger docs, and full pytest coverage (15/15 passing)
- Implemented MLOps: joblib model versioning, prediction logging, drift detection, and automated retraining trigger
- Containerized with Docker, orchestrated on Kubernetes with health checks, liveness probes, and autoscaling; CI/CD via GitHub Actions

Synapse Intelligence — Multi-Agent Business Research Assistant | Python, LangGraph, Google Gemini 2.0 Flash, Tavily MCP, Chainlit, Pydantic V2, SQLite, LangSmith

May 2026

- Built a 4-agent LangGraph pipeline (Clarity → Research → Validator → Synthesis) with conditional routing, retry loops, and human-in-the-loop interrupts for real-time company research
- Implemented stateful multi-turn conversations using AsyncSqliteSaver checkpointing and Pydantic V2 structured

outputs across all agent boundaries

- Integrated Tavily MCP for real-time web search with dual-search for comparison queries and direct JSON URL extraction to eliminate hallucinated sources
- Delivered a production-ready Chainlit web UI with token streaming, per-agent timing, conversation export, and switchable LLM providers (Gemini / Groq)

ReviewRoute — Multi-Model NLP Sentiment Analysis System | *Python, XLM-RoBERTa, HuggingFace, PyTorch, TF-IDF, scikit-learn, FastAPI* 2026

- Fine-tuned xlm-roberta-base on 60,000 multilingual Amazon reviews for 5-class star-rating classification using fp16 mixed precision and gradient checkpointing
- Built 3-model pipeline: TF-IDF + Logistic Regression baseline, fine-tuned transformer, and stacking meta-learner combining probability outputs and one-hot features
- Engineered smart routing engine detecting language, script type (CJK, Arabic, Cyrillic), and confidence-based escalation between models
- Deployed all models via FastAPI with Swagger docs, health checks, and Pydantic validation

RAG Chatbot — Retrieval-Augmented Generation Pipeline | *Python, LangChain, OpenAI API, Pinecone, FAISS, FastAPI* 2025

- Built end-to-end RAG pipeline with document ingestion, vector embedding (OpenAI), and hybrid retrieval using Pinecone (cloud) and FAISS (local)
- Served conversational AI interface via FastAPI with context-aware prompt engineering and response streaming

AI Connect 2025 — International CSP Solver Competition | *Python, CSP, Backtracking, MRV, AC-3, HuggingFace Datasets* 2025

- Represented NUST in 11-day international AI challenge with universities from Germany, Turkey, Pakistan, and Bhutan
- Built hybrid CSP solver for ZebraLogicBench (~1,000 logic puzzles) using backtracking search with MRV heuristic, Forward Checking, and Arc Consistency (AC-3)
- Optimized composite score balancing accuracy (%) and search-step penalty, achieving both correctness and efficiency

ACHIEVEMENTS

- Fine-tuned XLM-RoBERTa (Meta) on 60,000 multilingual reviews with production-grade training configuration (fp16, gradient checkpointing, gradient accumulation)
- Built production multi-agent AI system with **LangGraph** — 4 specialized agents, human-in-the-loop interrupts, and real-time web search via Tavily MCP
- International AI competitor — AI Connect 2025, co-hosted by HSBI Germany, TDU Turkey, NUST Pakistan, CST/RUB Bhutan
- Improved database query speed by 30% at Rewild Mobile through schema optimization
- Reduced booking errors by 25% through frontend validation in Sports Complex Management System
- Delivered 3+ major projects ahead of deadlines across academic and professional settings